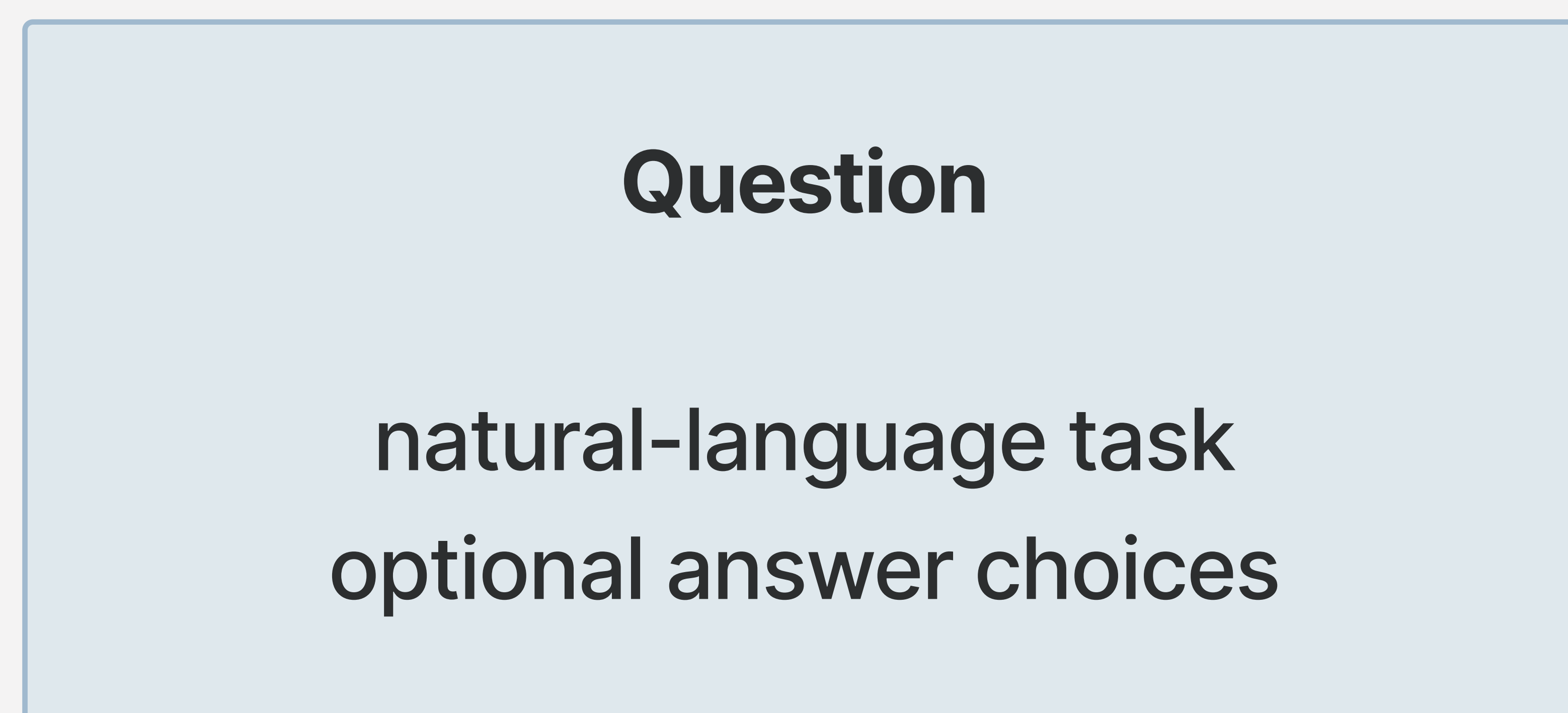
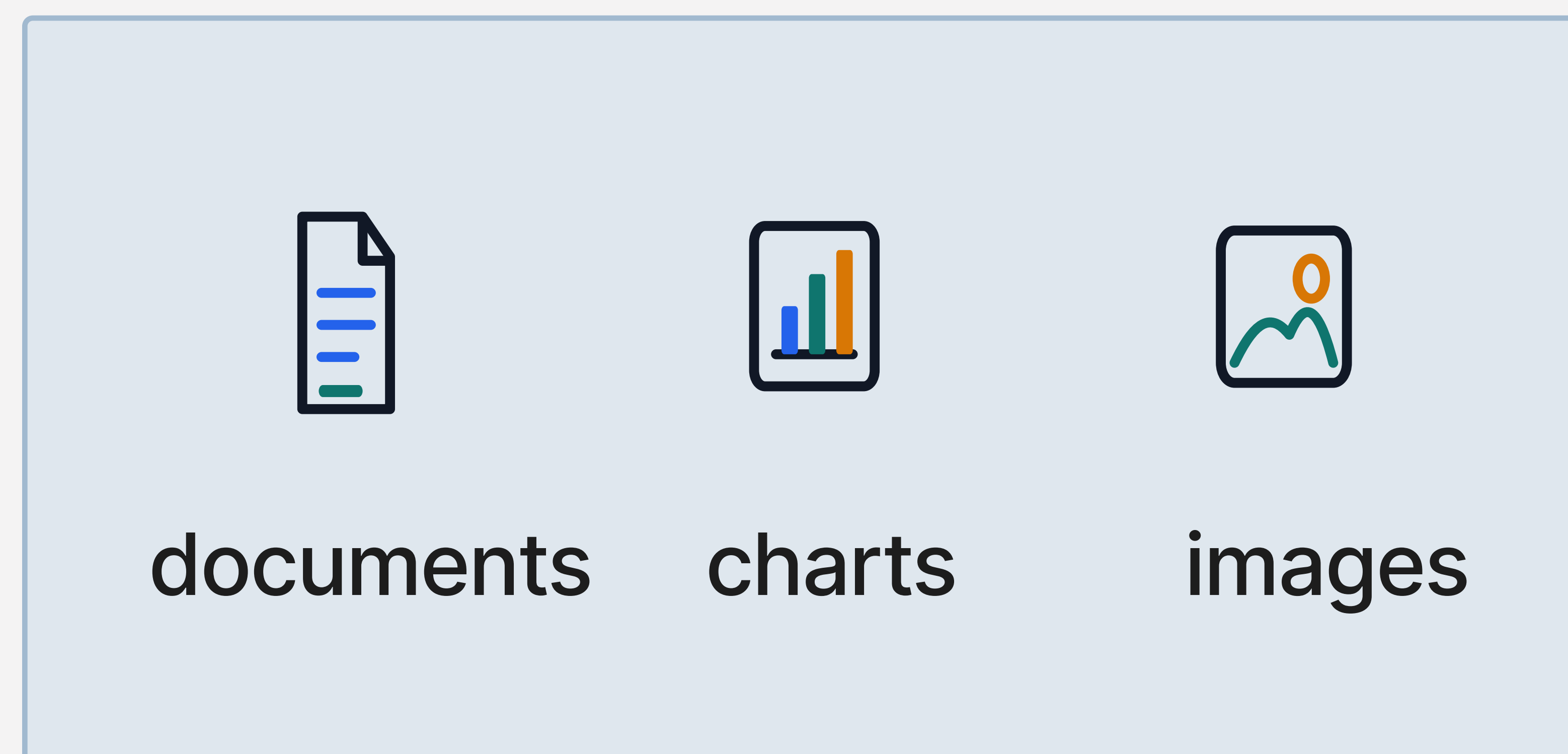


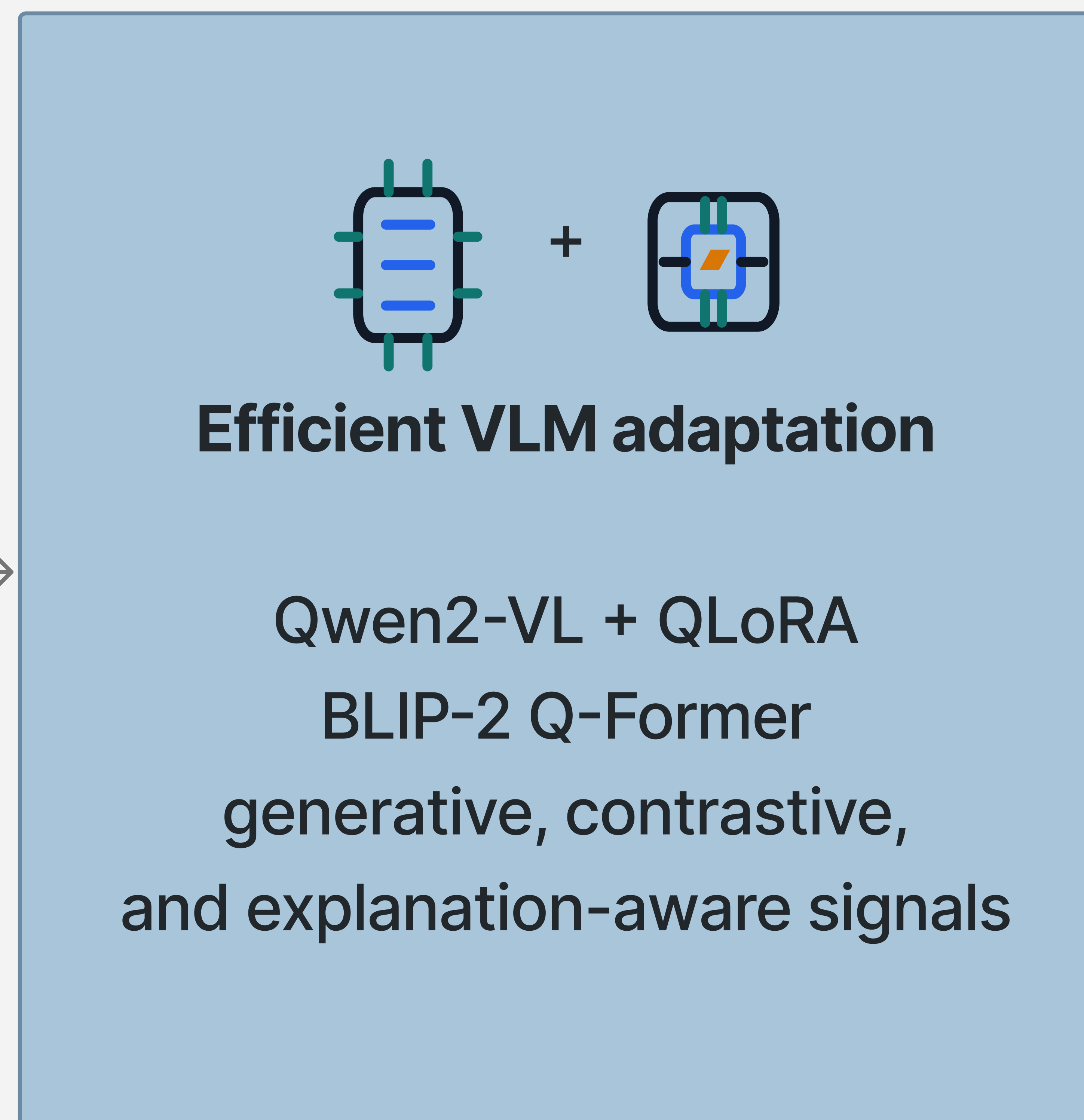
Controlled explanation-aware VLM alignment study

fixed-budget comparison of generative, contrastive, and explanation-aware supervision

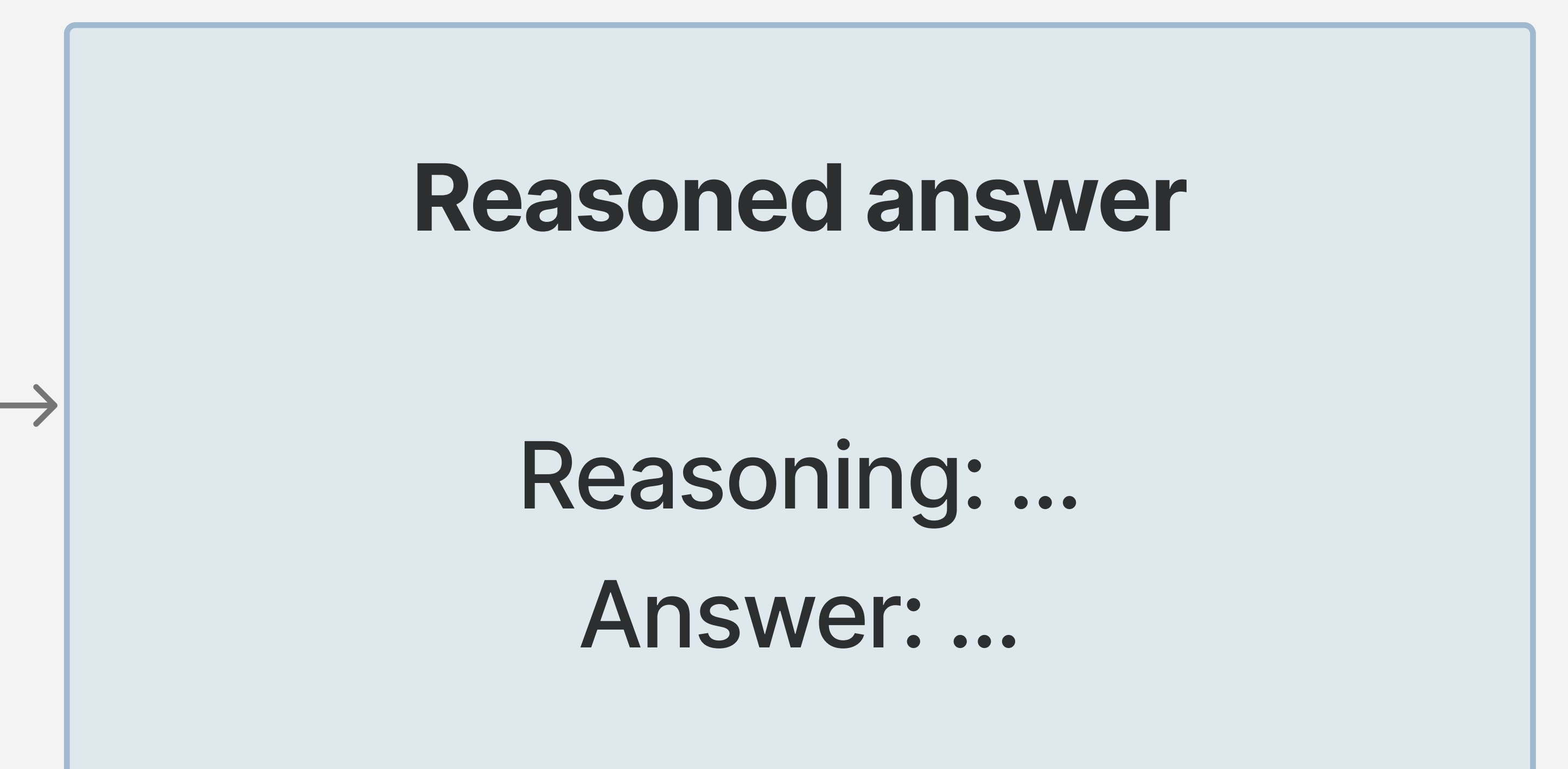
visual inputs



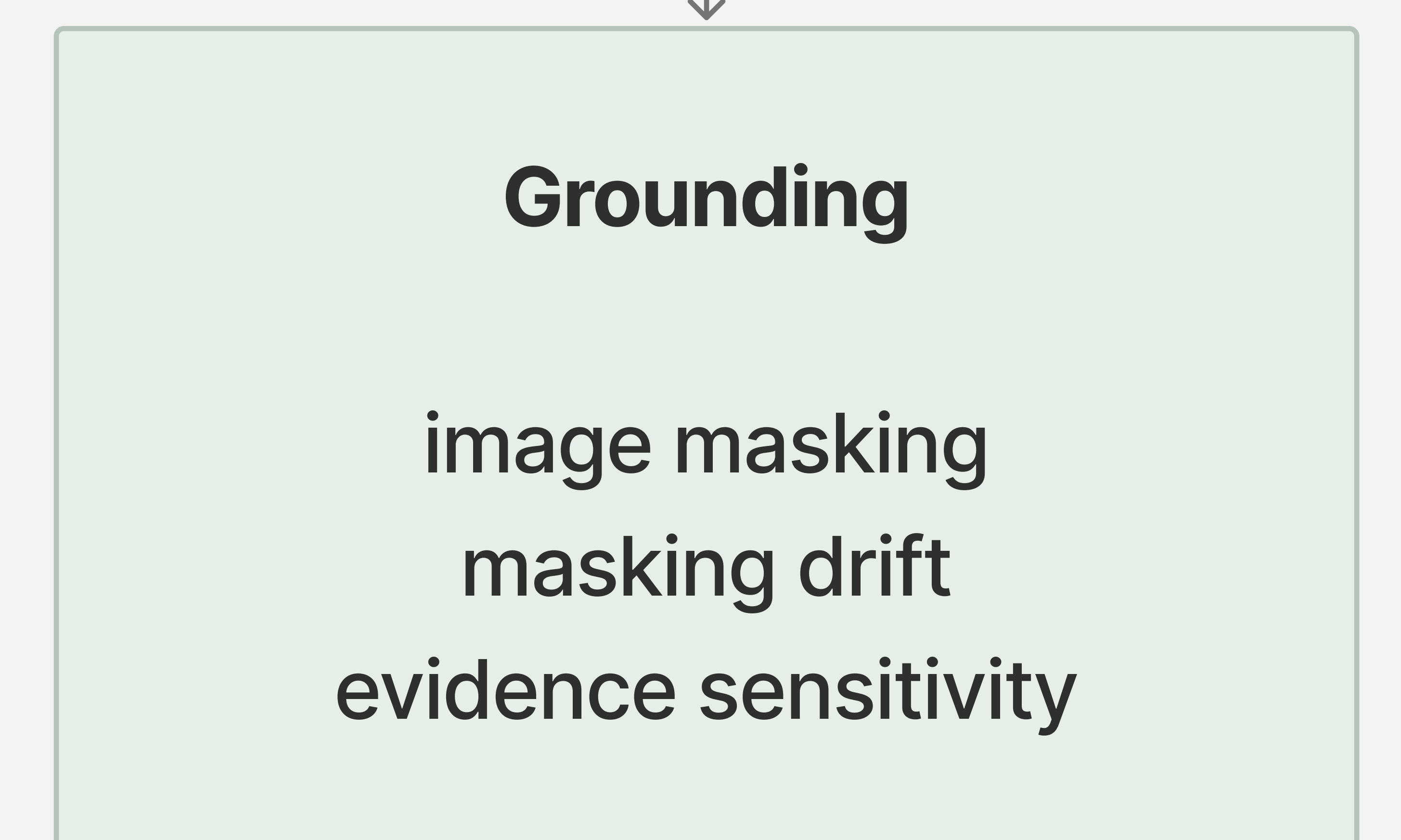
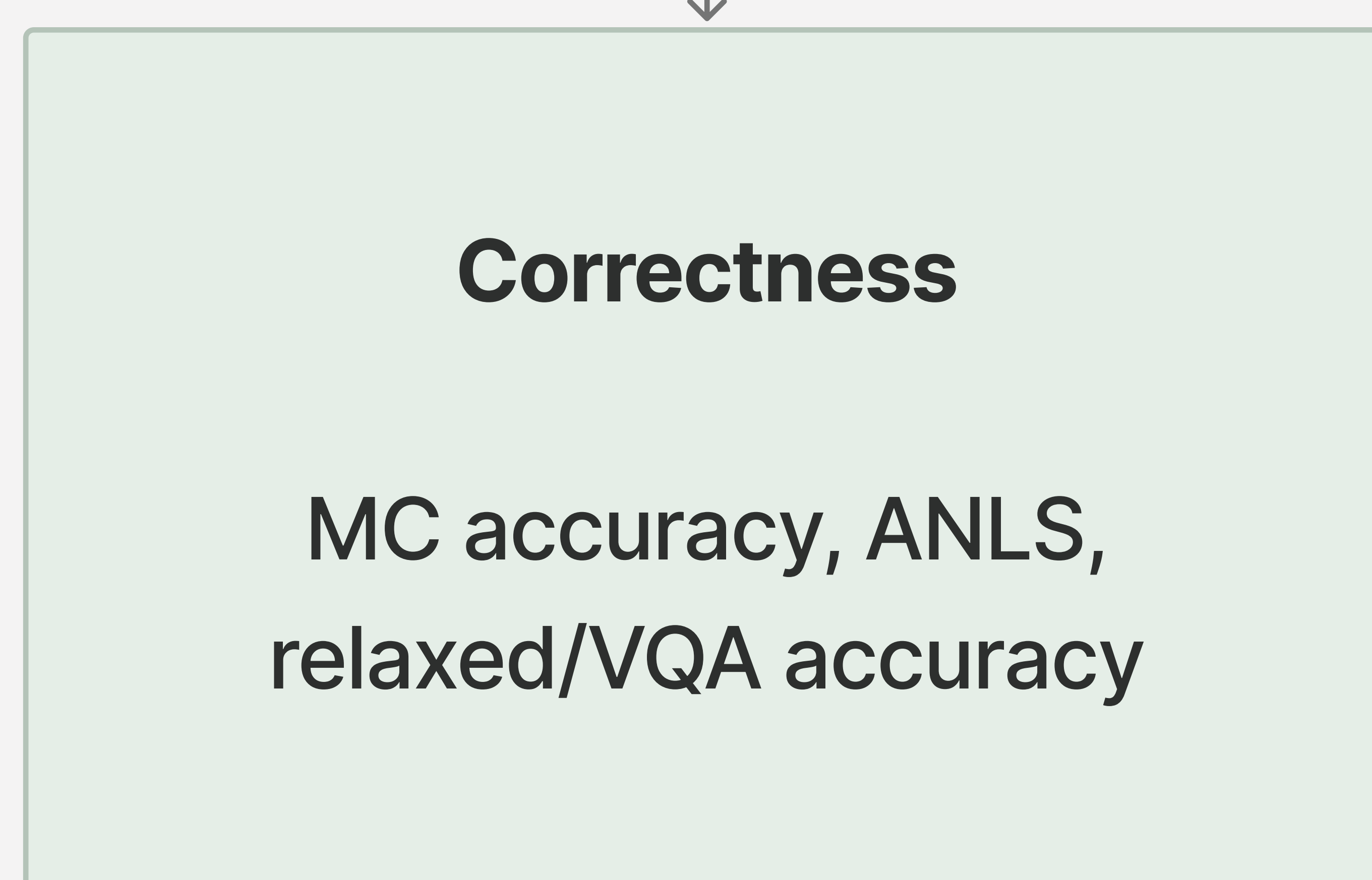
alignment signal



generation



evaluation axes



Controlled study: same budget, same protocol, only the training signal changes.